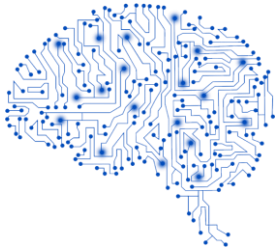
Recommendation System

Basic Concepts

Hoang-Quynh Le (PhD), VNU-UET



Outline

- Real-life Examples
- Introduction to Recommendation Systems (RSs)
  - Item, User, Behavior
  - Role of RSs ? How do RSs work?
- Types of Recommendation Systems
- Learning Principles
  - Learning User-Item Associations
  - Learning Item-Item Associations
- Evaluation & Metrics
- Introduction to Cornac, a Recommendation Framework

1.

Real-life Examples

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Help us show you better ads by updating your ads preferences.
Google
3 Sick Bodybuilding Tips
by sixpackshortcuts
1,602,424 views
3:32
Ad
Ab To Hai Tum Sa Her Khushi Apni lata
by malikje1
45,661
FEATURED
4:00
Pink Interview on her husband and daughter
by MsSupergalaxy
Recommended for you
12:44
Abhimaan (1973): Loote Koi Man Ka Nagar
by tonguesonfire
5:56 / 9:55
Ads by Google
Song:Tere Mere Milan Ki Yeh Raina Film: Abhimaan (1973) With Sinhala S...
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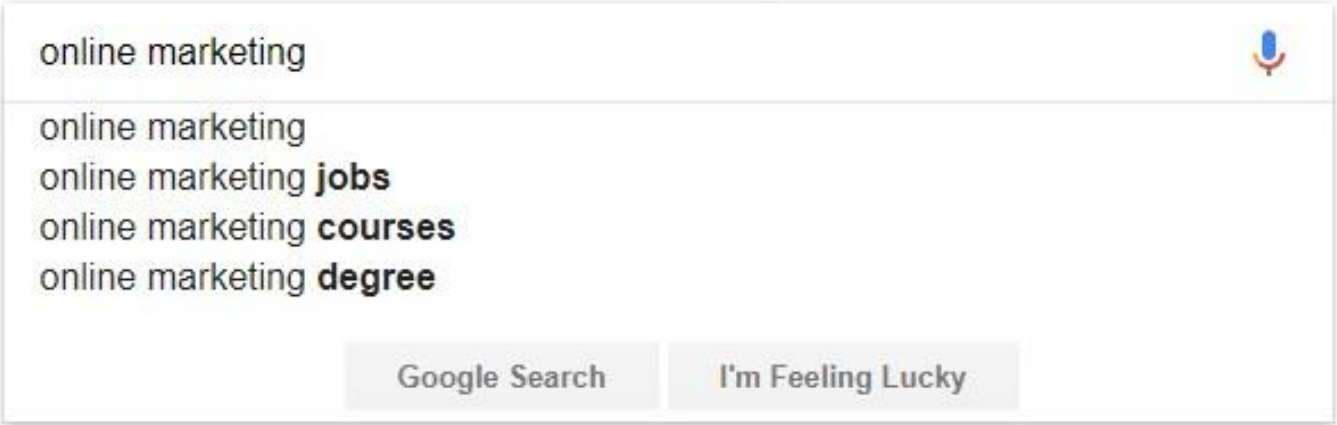
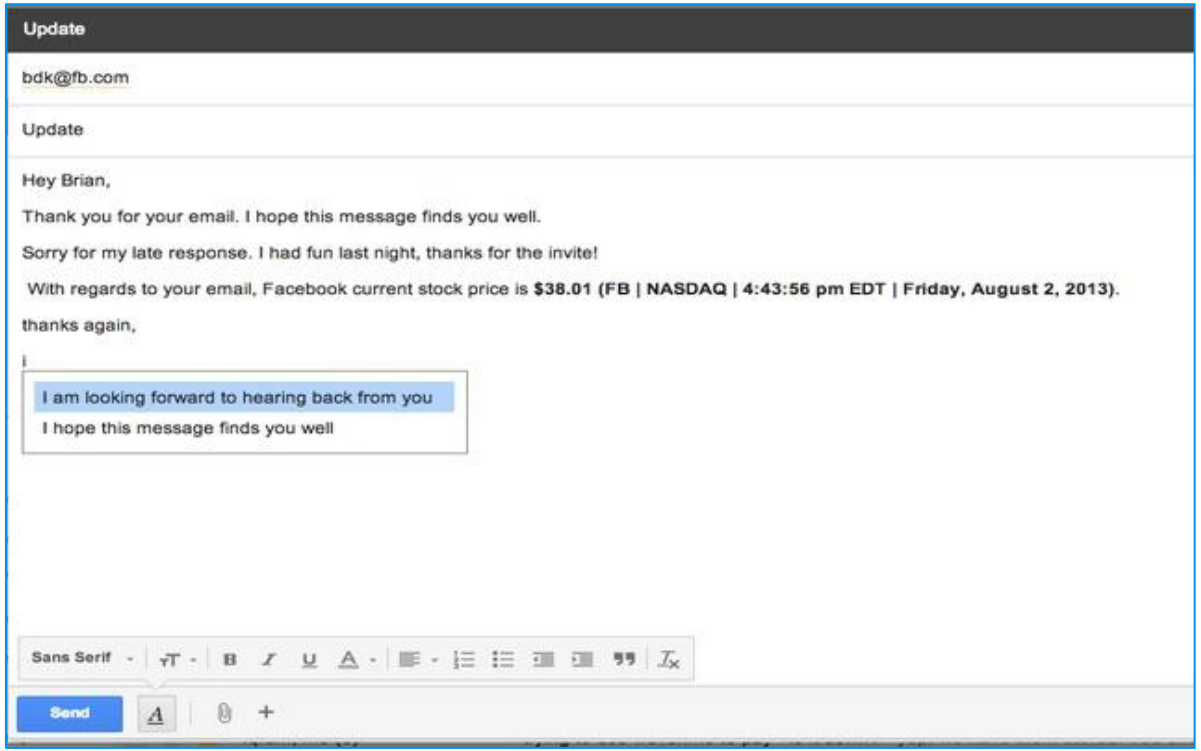


Gmail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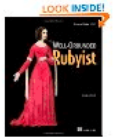

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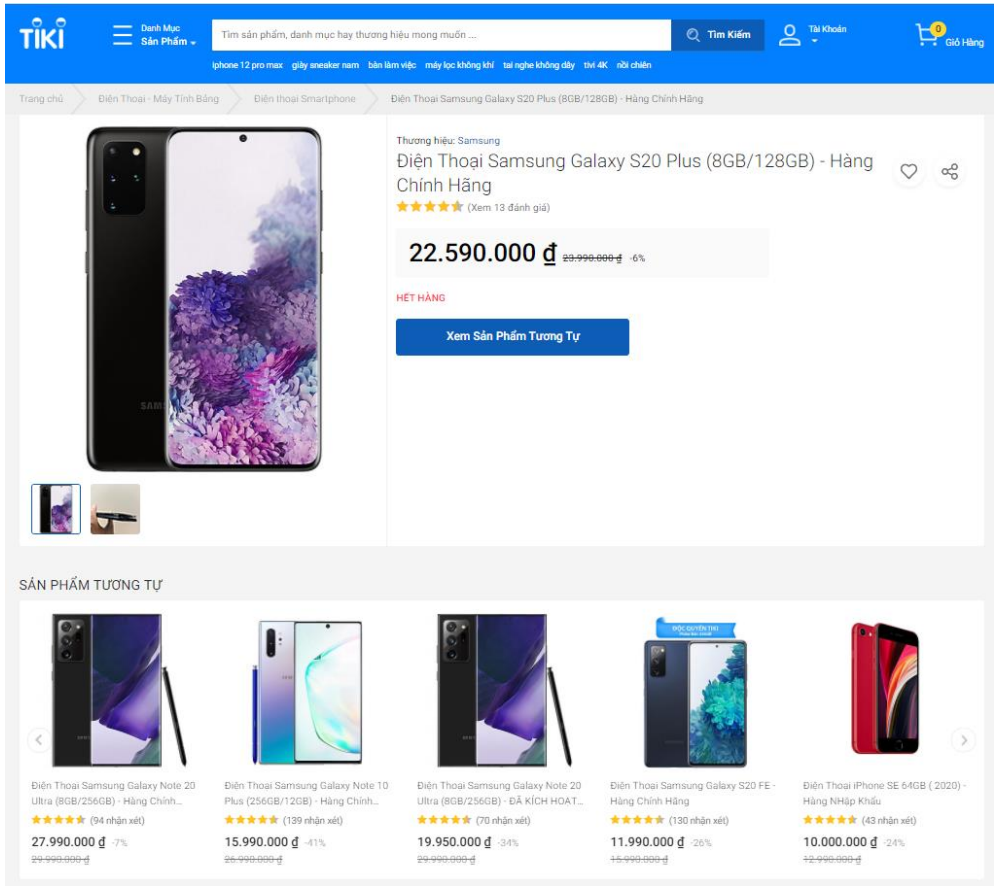
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Danh Mục Sản Phẩm
Tìm sản phẩm, danh mục hay thương hiệu mong muốn ...
Tìm Kiếm
Tài Khoản
Giỏ Hàng
Iphone 12 pro max giày sneaker nam bàn làm việc máy lọc không khí tai nghe không dây tivi 4K nồi chiên
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Điện Thoại - Máy Tính Bảng
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Thương hiệu: Samsung
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(Xem 13 đánh giá)
22.590.000 ₫
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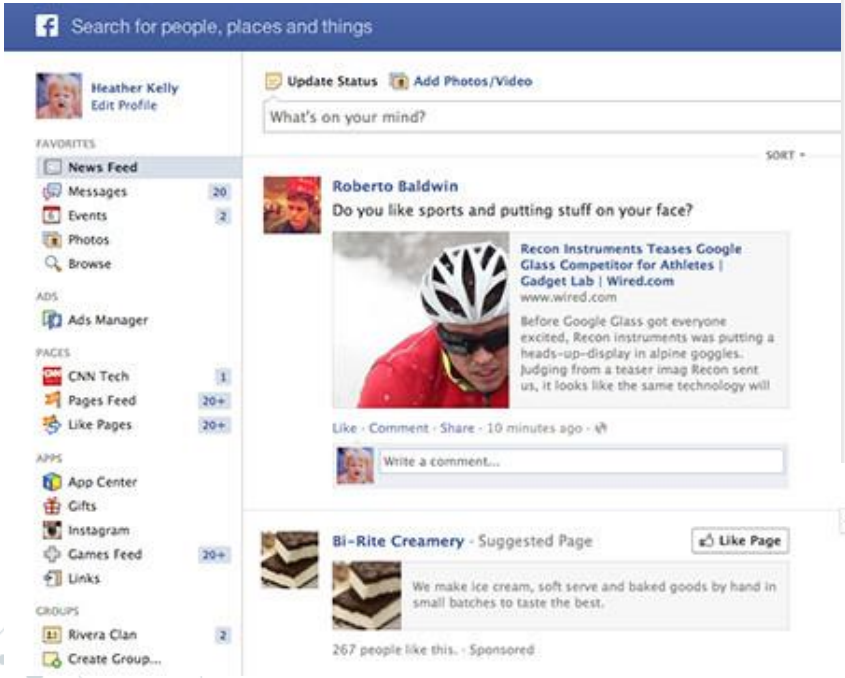
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PAGES
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Like Pages 20+
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Gifts
Instagram
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Update Status
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SORT
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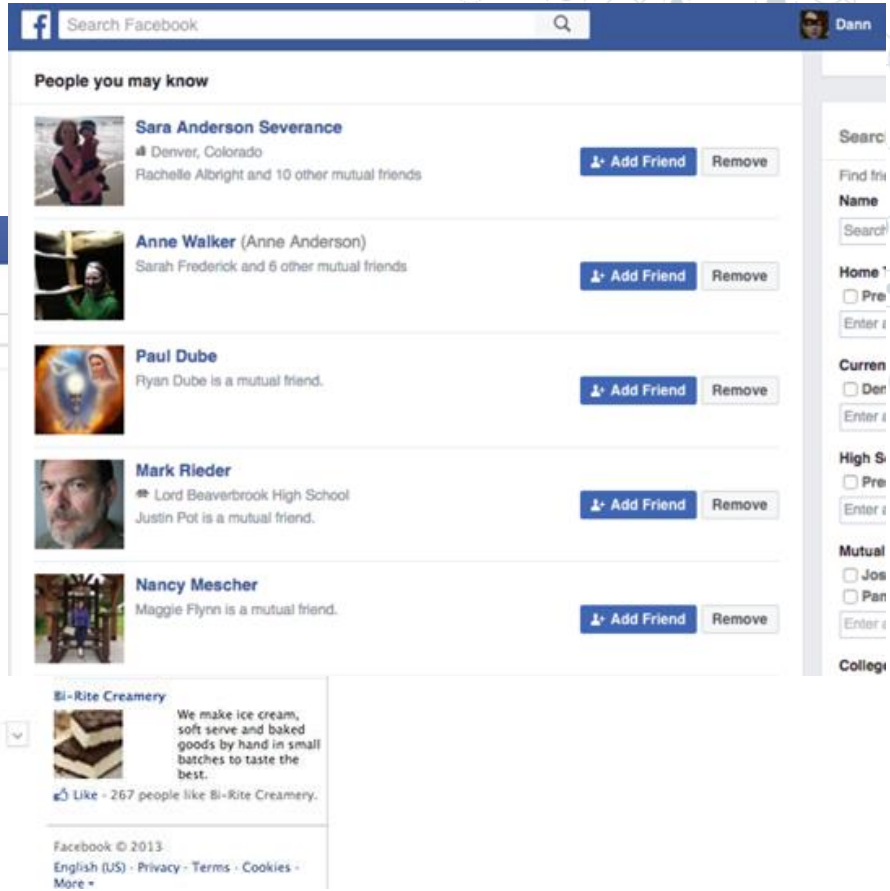
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Paul Dube
Ryan Dube is a mutual friend.
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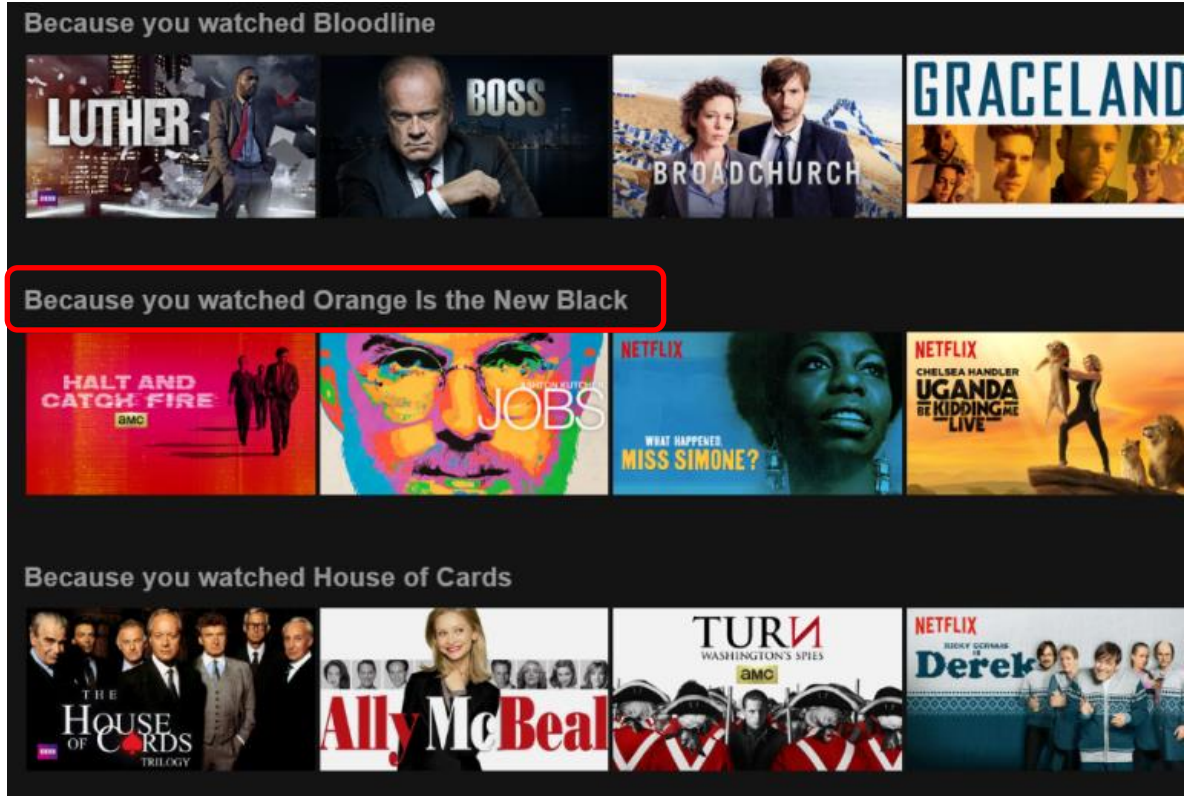
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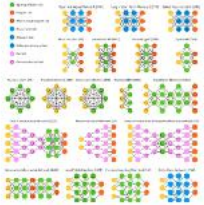

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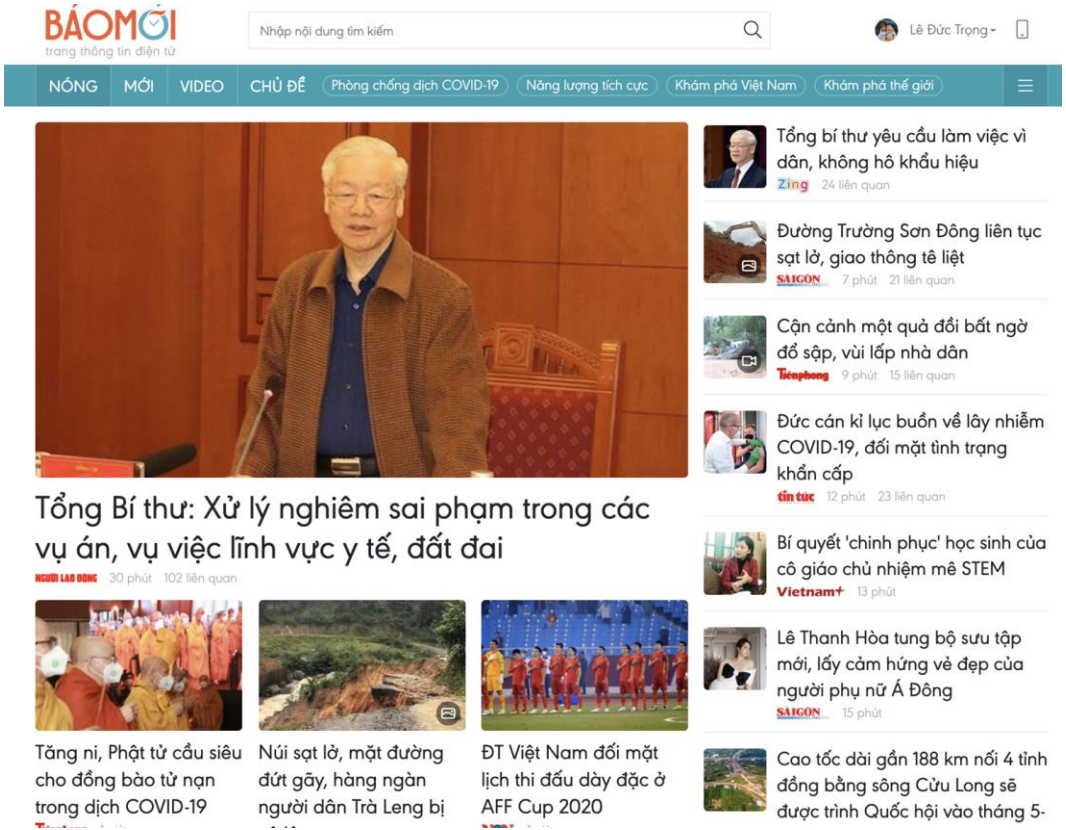
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Năng lượng tích cực
Khám phá Việt Nam
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Tổng Bí thư: Xử lý nghiêm sai phạm trong các vụ án, vụ việc lĩnh vực y tế, đất đai
NGƯỜI LAO ĐỘNG 30 phút 102 liên quan
Tăng ni, Phật tử cầu siêu cho đồng bào tử nạn trong dịch COVID-19
Núi sạt lở, mặt đường đứt gãy, hàng ngàn người dân Trà Leng bị
ĐT Việt Nam đối mặt lịch thi đấu dày đặc ở AFF Cup 2020
Tổng bí thư yêu cầu làm việc vì dân, không hô khẩu hiệu
Zing 24 liên quan
Đường Trường Sơn Đông liên tục sạt lở, giao thông tê liệt
SÀI GÒN 7 phút 21 liên quan
Cận cảnh một quả đồi bất ngờ đổ sập, vùi lấp nhà dân
Tiềnphong 9 phút 15 liên quan
Đức cán kỉ lục buồn về lây nhiễm COVID-19, đối mặt tình trạng khẩn cấp
tin tức 12 phút 23 liên quan
Bí quyết 'chinh phục' học sinh của cô giáo chủ nhiệm mê STEM
Vietnam+ 13 phút
Lê Thanh Hòa tung bộ sưu tập mới, lấy cảm hứng vẻ đẹp của người phụ nữ Á Đông
SÀI GÒN 15 phút
Cao tốc dài gần 188 km nối 4 tỉnh đồng bằng sông Cửu Long sẽ được trình Quốc hội vào tháng 5-

Recommendations everywhere

2.

Introduction to Recommendation Systems

Item, User & Behavior

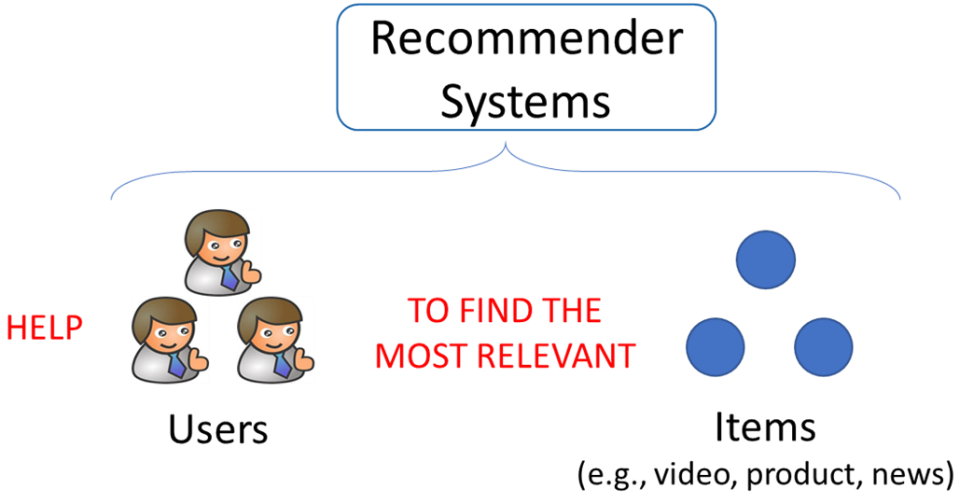
Customer A

	Behavior	Item
YouTube	Click, Like/Dislike, Comment, Search	**Video**
amazon	Search, Click, Add-to-cart, Purchase, Review, Rate	**Product**
NETFLIX	Search, Click, Rate, Like	**Movie**
BáoMới.com	Search, Read, Duration, Comment	**News**
...	...	...

Why do we need Recommendation?

Millions of Items ...

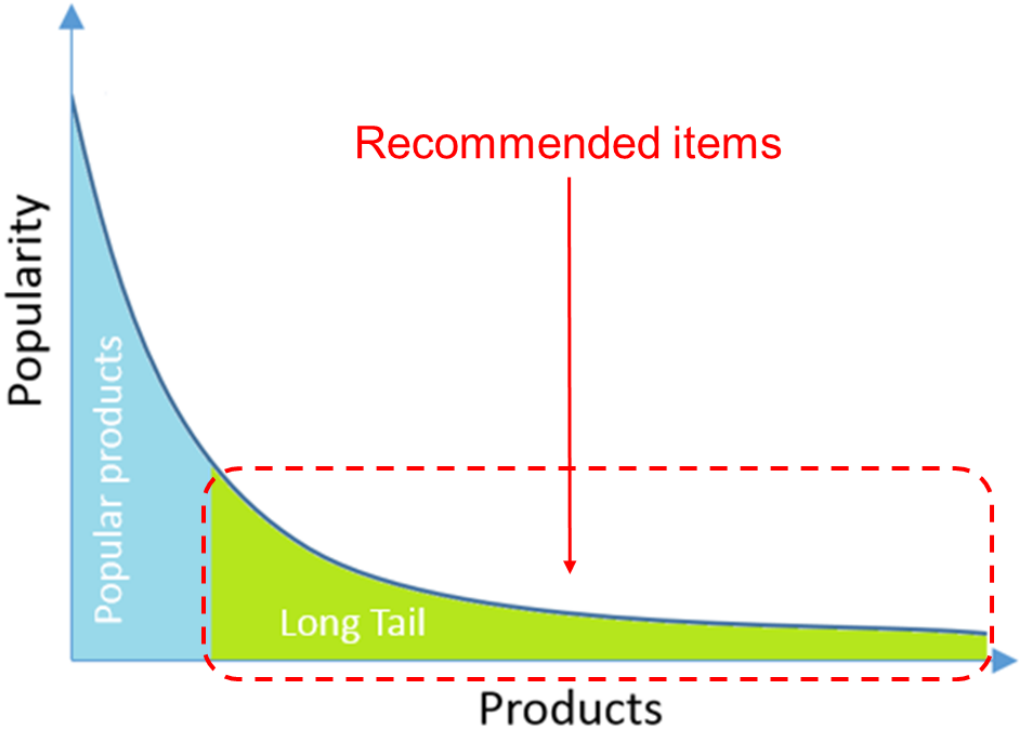
Roles of Recommendation Systems



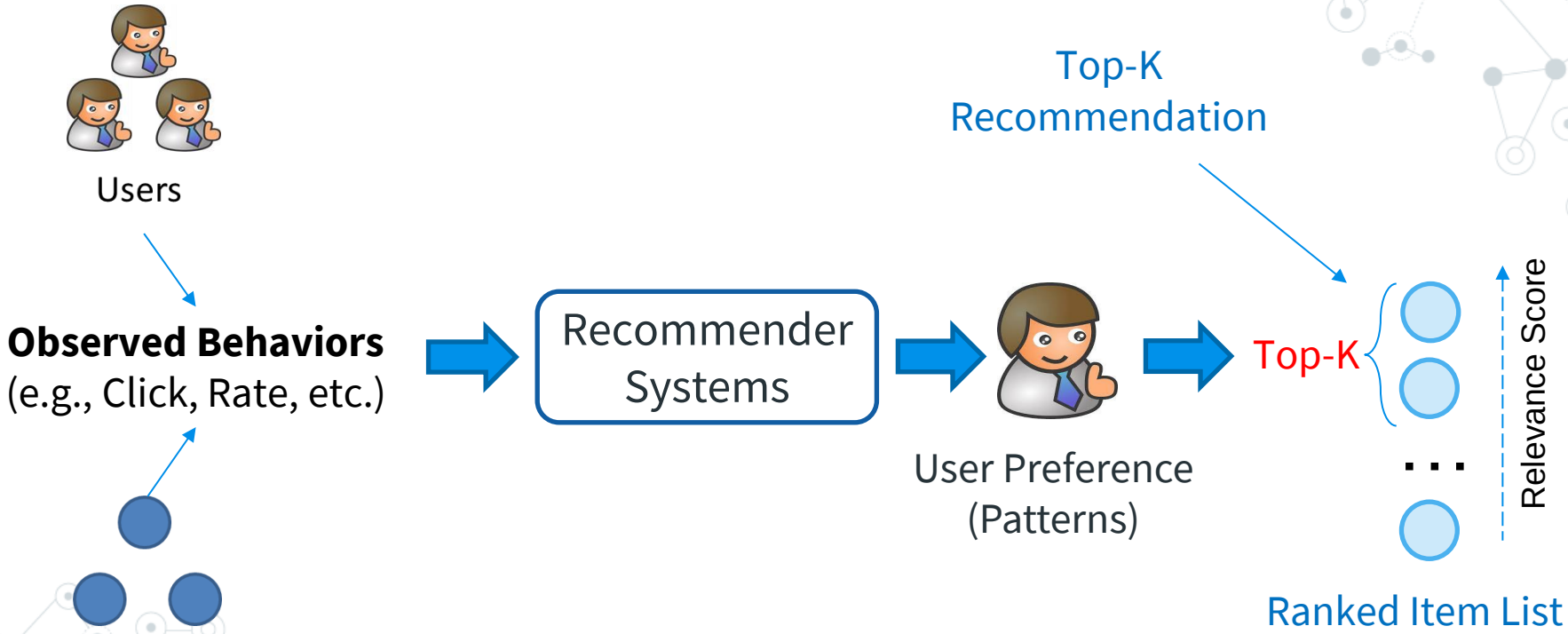
Advantages of RS:

Enterprise Perspective	Customer Perspective
• Reduce search costs • Increase customer satisfaction • Educate customers about product domains. • Optimize sales and profit	• Easy to find what he/she might want • Having an assistant in websites/systems

Long-tail Recommendation



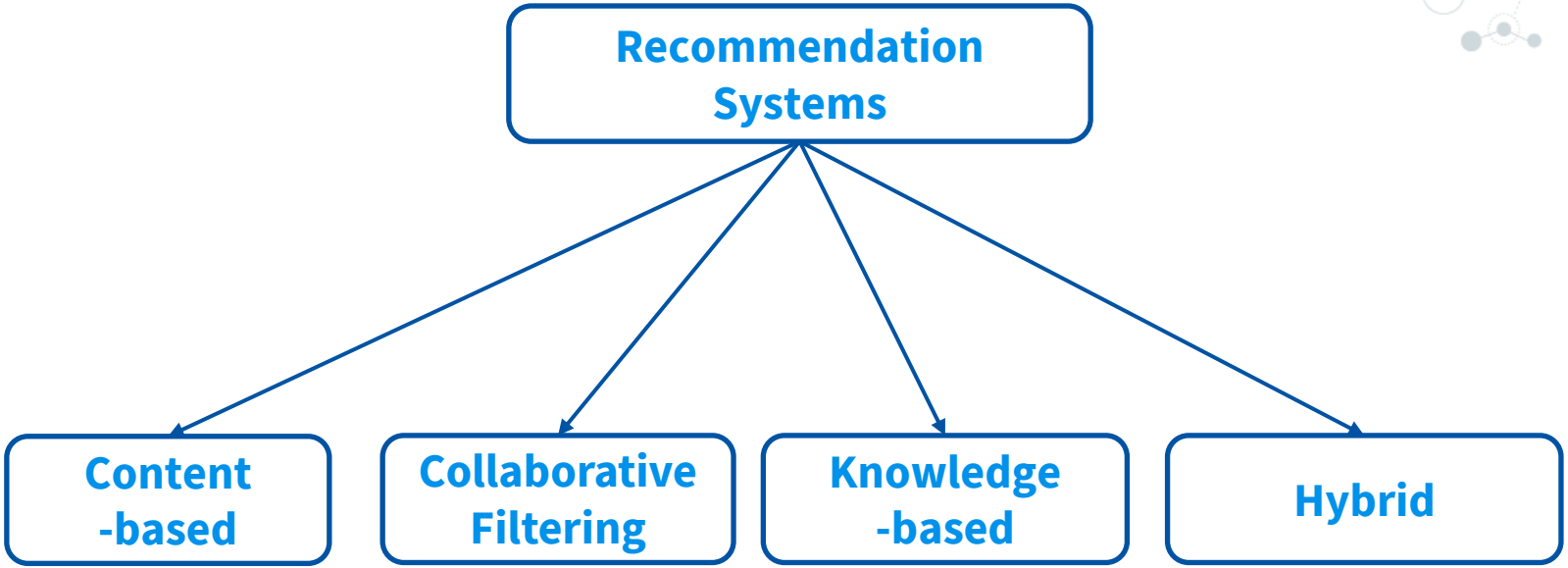
How does RS work?



3.

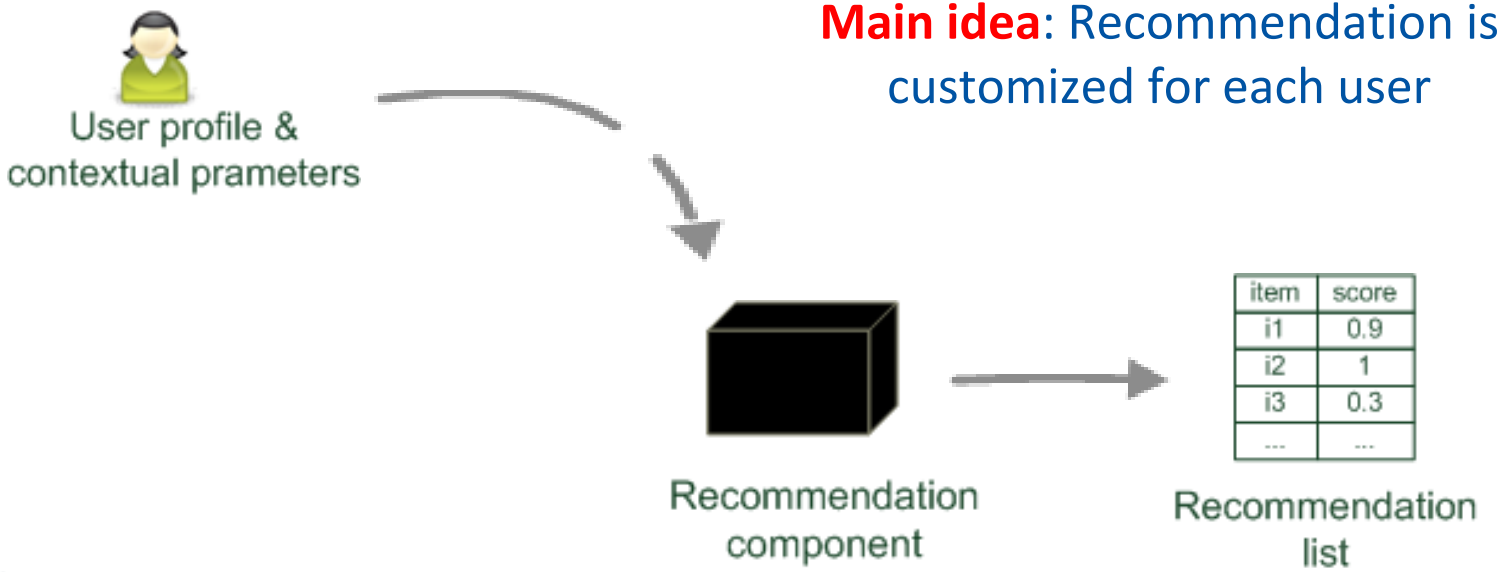
Types of Recommendation Systems

Types of Recommendation Systems



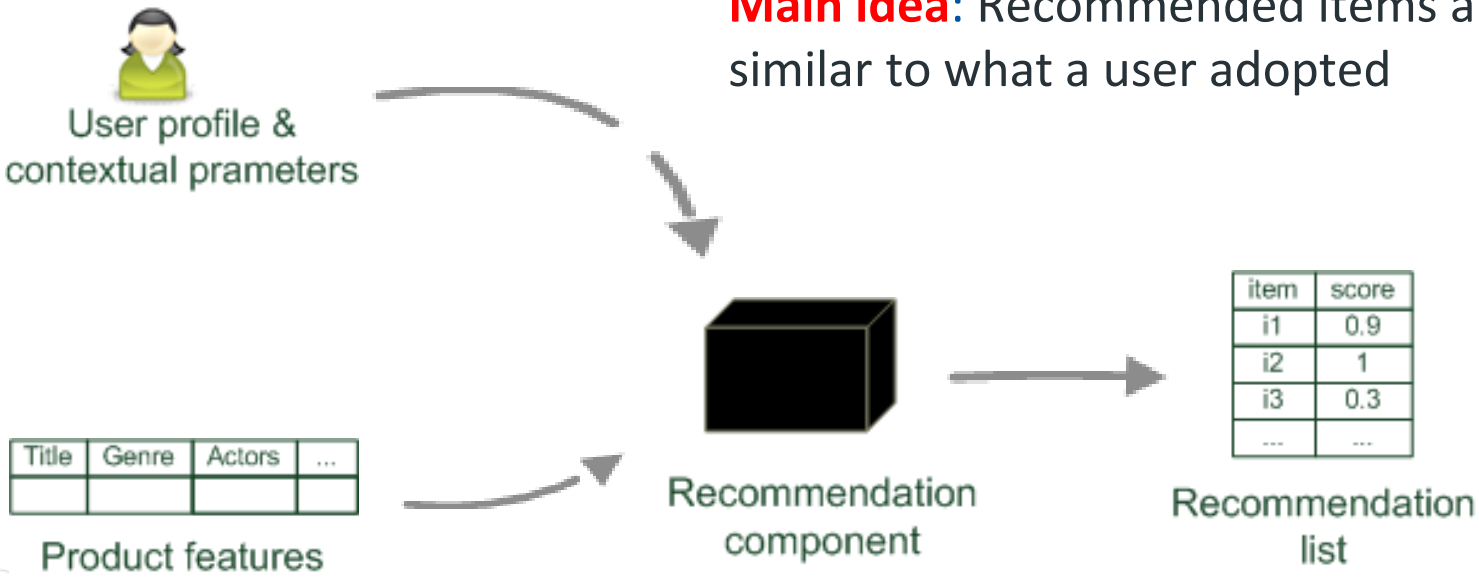
Personalized Recommendation

Main idea: Recommendation is customized for each user



Content-based Recommendation (1)

Main idea: Recommended items are similar to what a user adopted



Content-based Recommendation (2)

WD Blue 1TB SATA 6 Gb/s 7200 RPM 64MB Cache 3.5 Inch Desktop Hard Drive (WD10EZEX)

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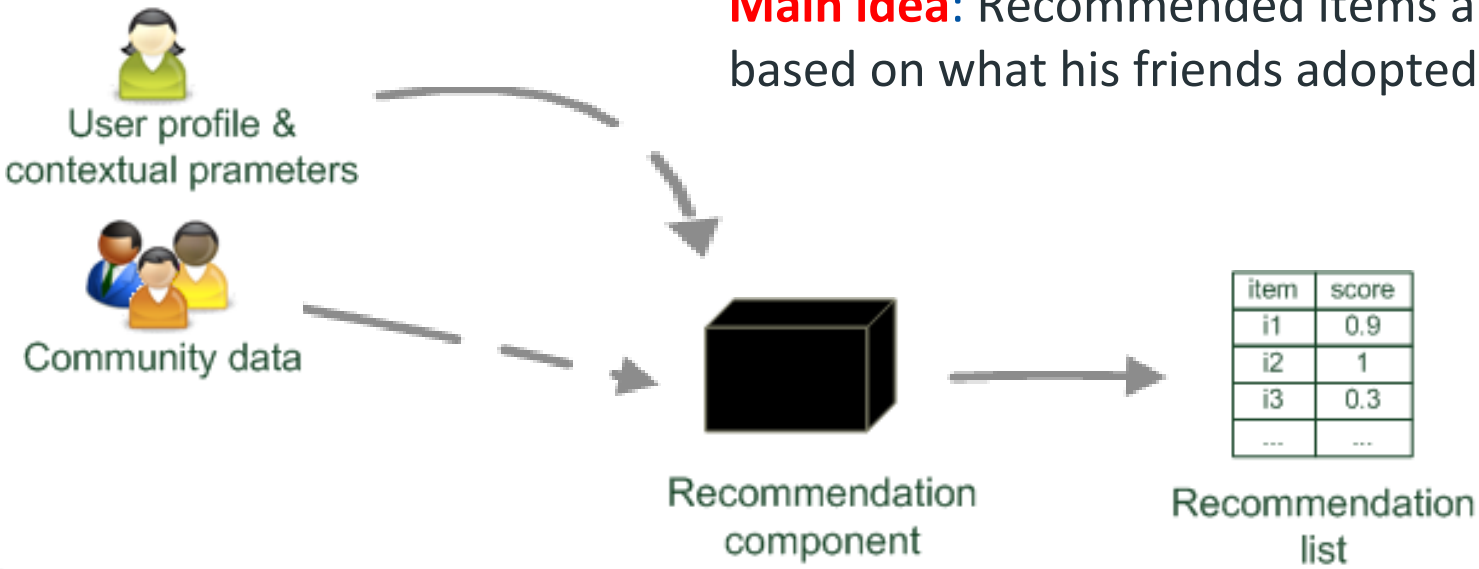
WD Black 1TB Performance Desktop Hard Disk Drive - 7200 RPM SATA 6 Gb/s 64MB Cache 3.5 Inch - WD1003FZEX

Add to Cart

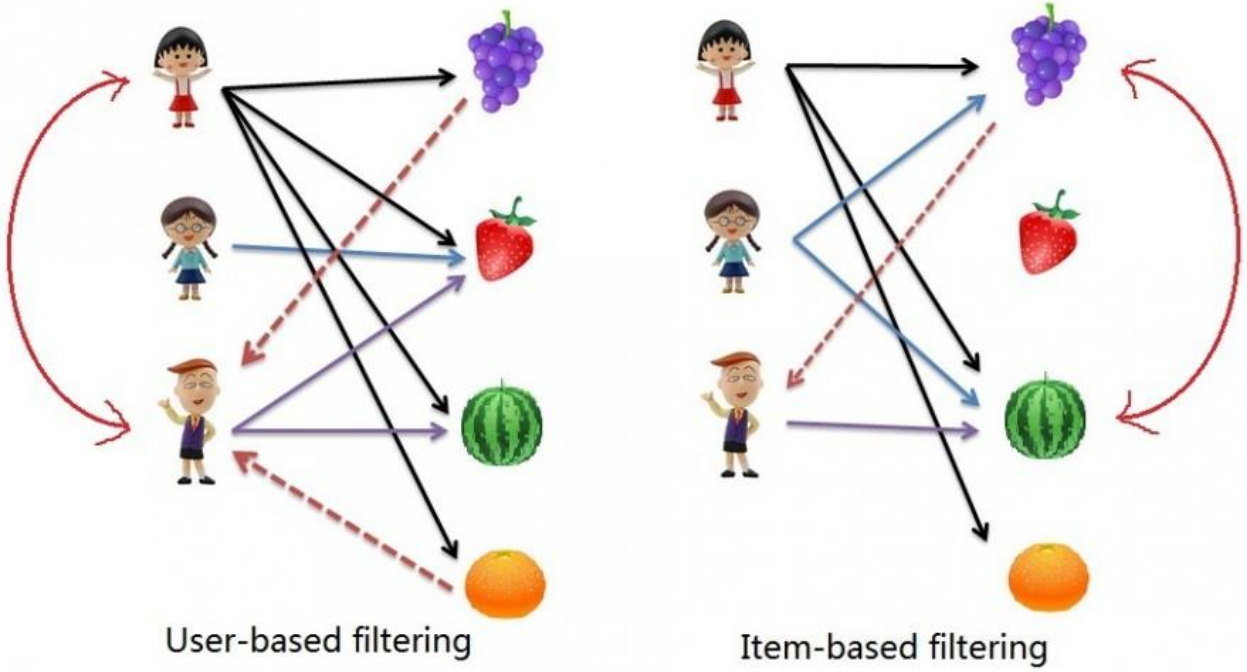
Recommendations for 1TB HDD on Amazon

Collaborative Filtering (1)

Main idea: Recommended items are based on what his friends adopted



Collaborative Filtering (2)



Netflix Prize

Leaderboard

Showing Test Score. Click here to show quiz score

Display top 20 leaders.

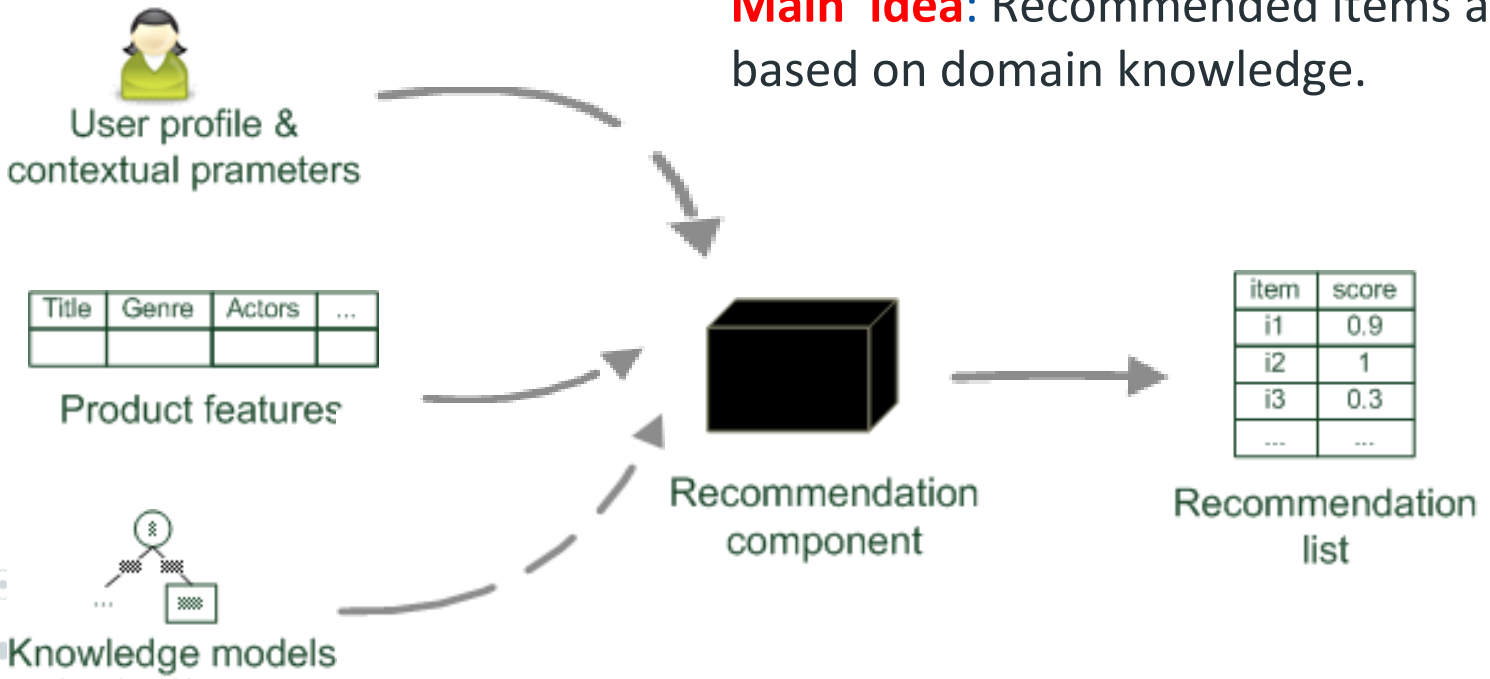
Rank	Team Name	Best Test Score	% Improvement	Best Submit Time
Grand Prize - RMSE = 0.8567 - Winning Team: BellKor's Pragmatic Chaos				
1	BellKor's Pragmatic Chaos	0.8567	10.06	
2	The Ensemble	0.8567	10.06	
3	Grand Prize Team	0.8582	9.90	
4	Opera Solutions and Vandelay United	0.8588	9.84	
5	Vandelay Industries !	0.8591	9.81	
6	PragmaticTheory	0.8594	9.77	
7	BellKor in BigChaos	0.8601	9.70	
8	Dace	0.8612	9.59	

	Movie 1	Movie 2	Movie 3
Ted	4	5	5
Carol		5	5
Bob		5	?

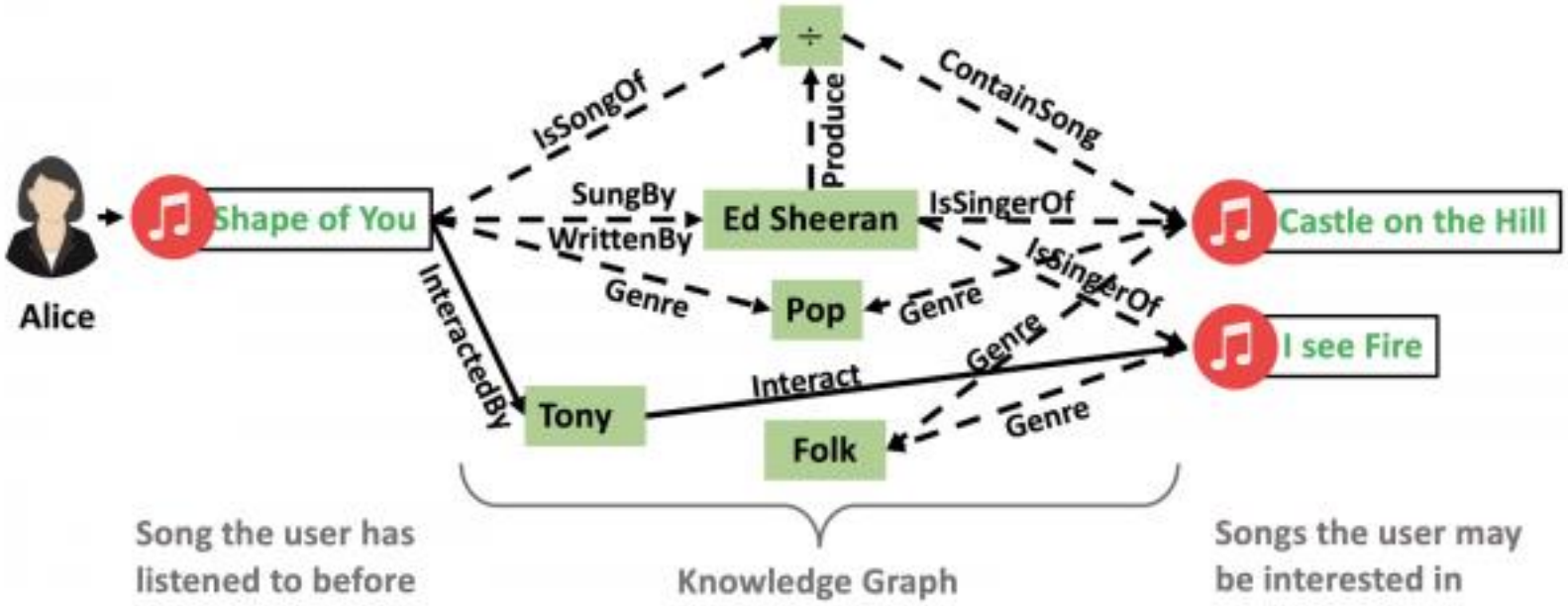
Netflix 1M $ challenge

Knowledge-based Recommendation (1)

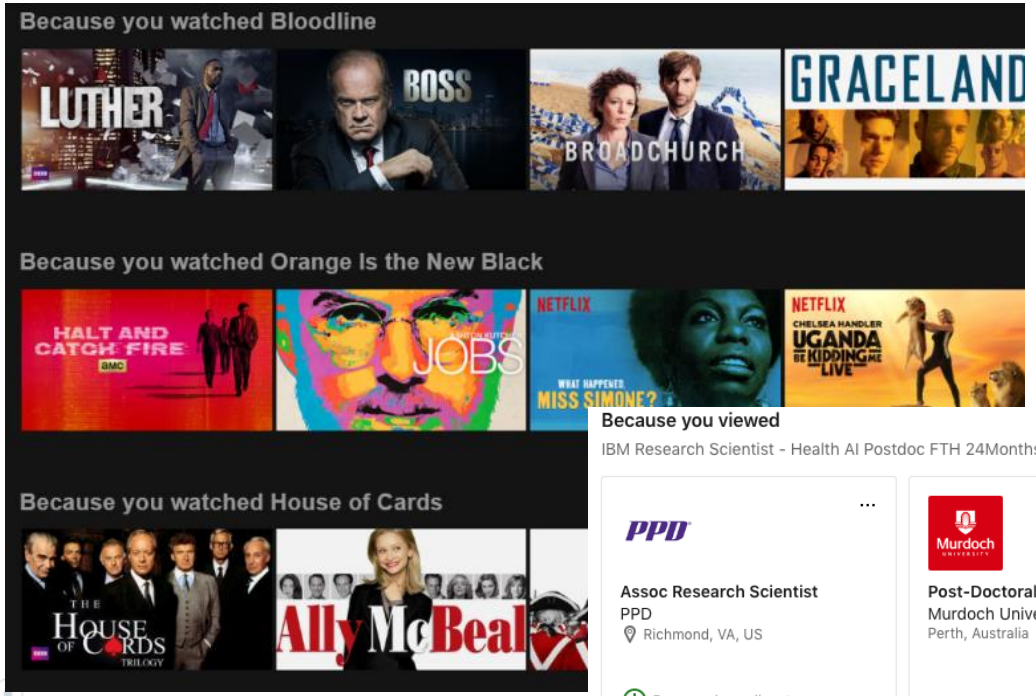
Main idea: Recommended items are based on domain knowledge.



Knowledge-based Recommendation (2)



Knowledge-based Recommendation (3)



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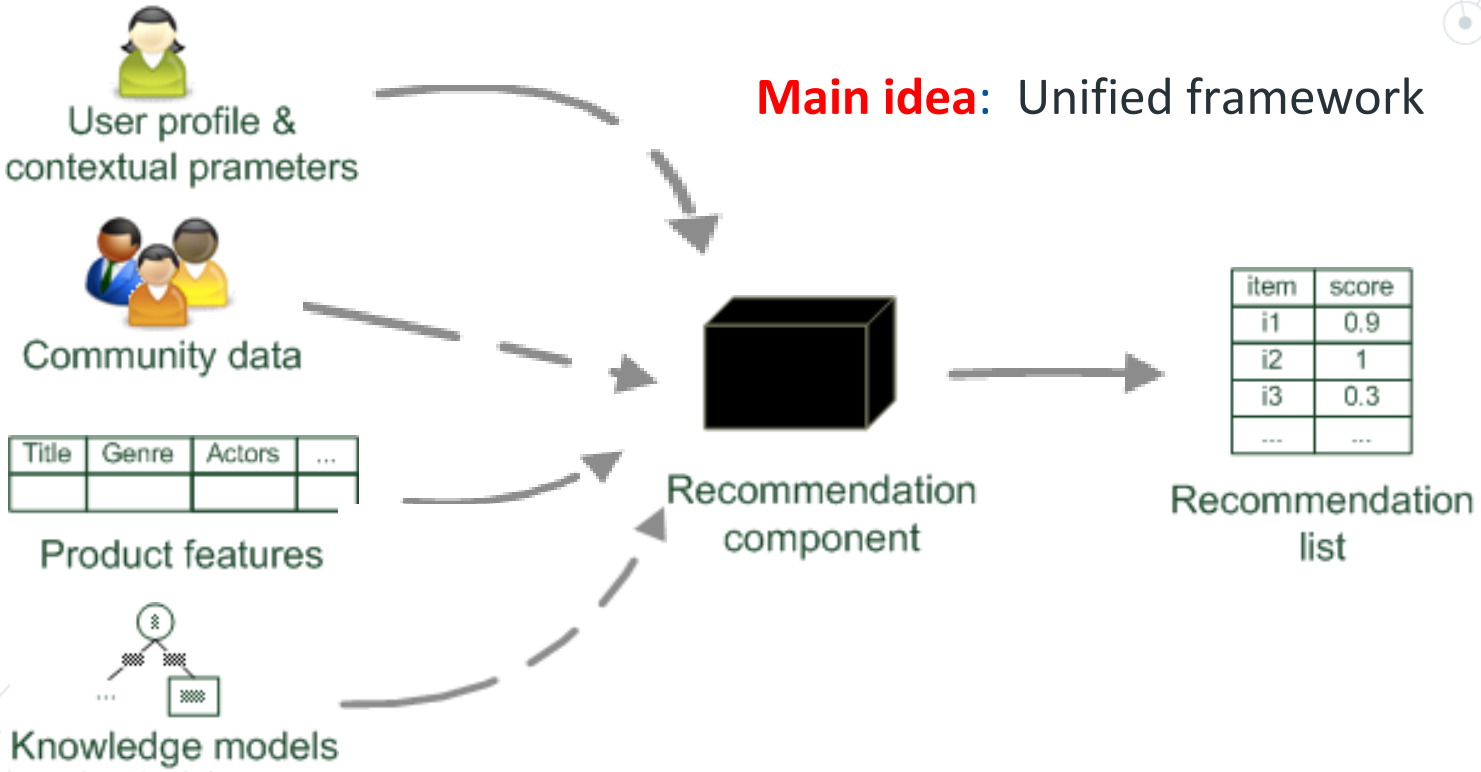
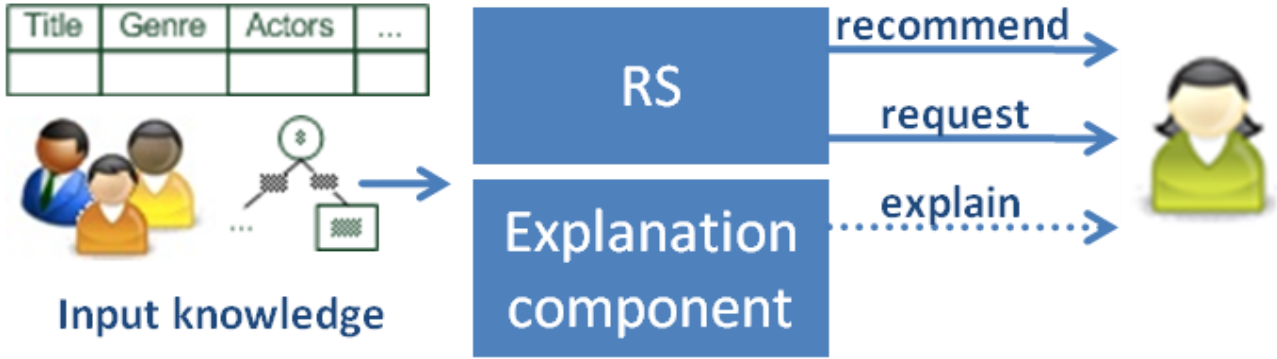
Hybrid Recommendation



Main idea: Unified framework

Explainable Recommendation

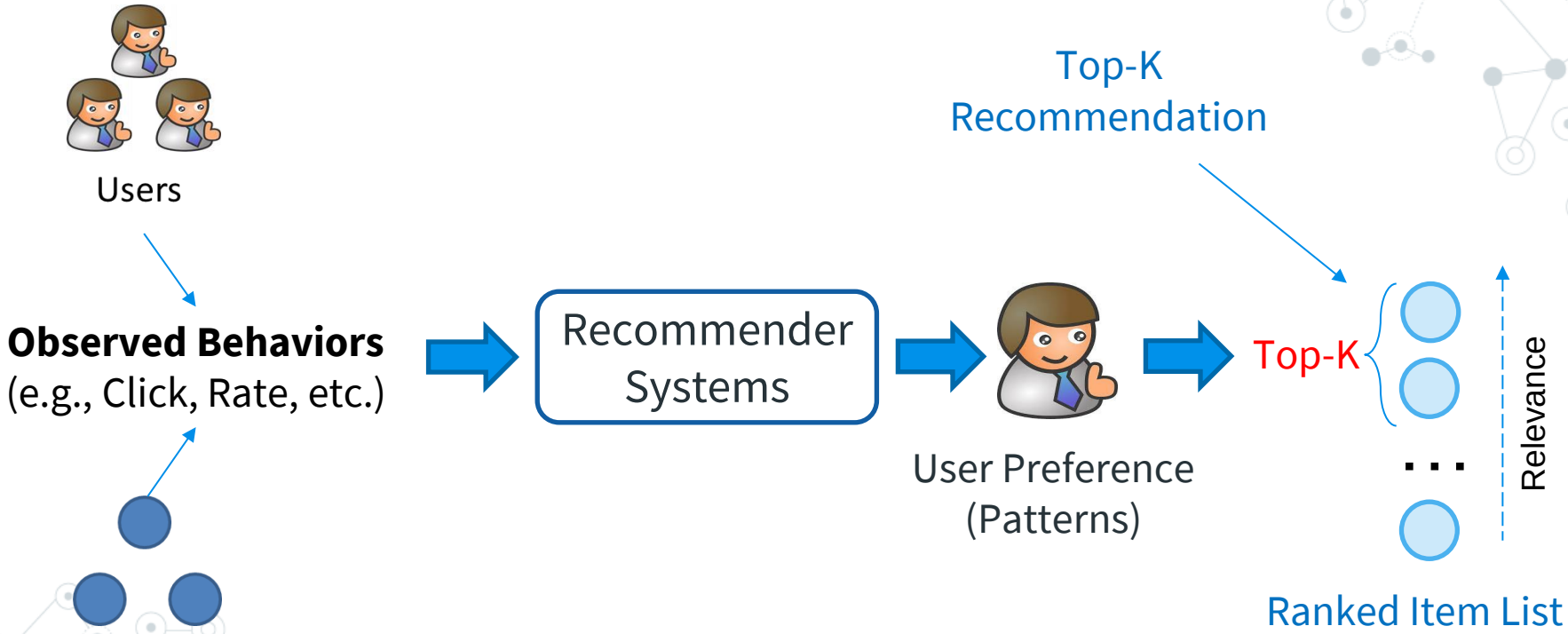


Main idea: Recommendation with explanation

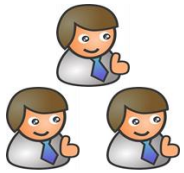
4.

Learning Principles

How does RS work?



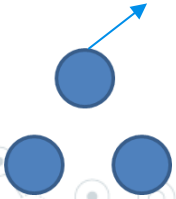
Top-K Recommendation



Users

<u>Main idea</u>: Learning **user preferences** via modeling **associations** to generate **top-K recommendations**.

Observed Data:
User-Item & Item-Item Associations

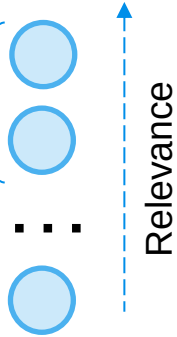


Items

Recommender Systems

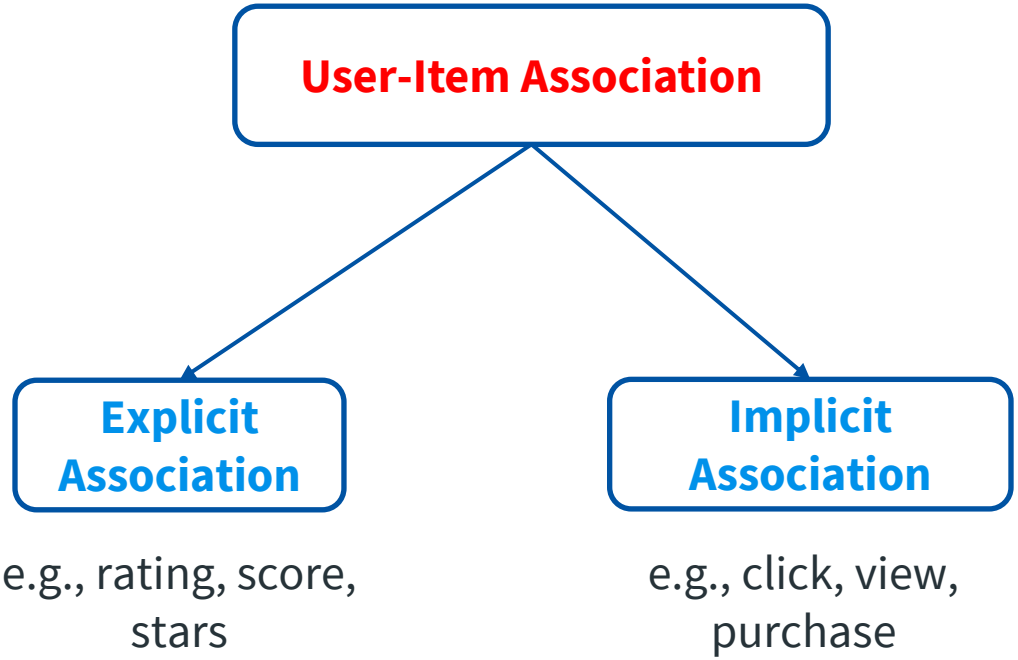
User Preference

Top-K

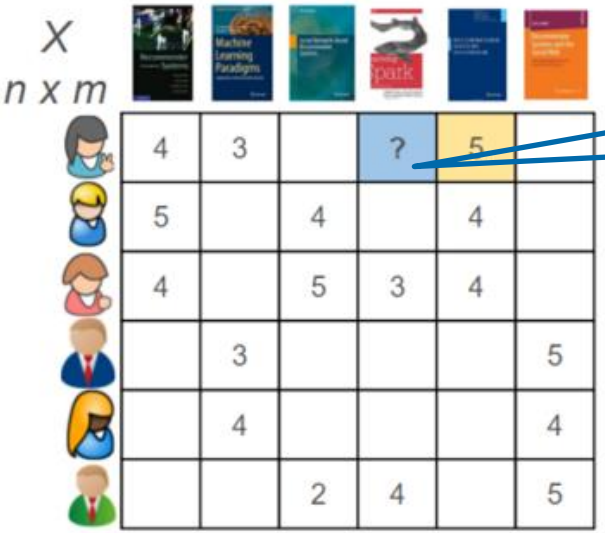


Ranked Item List

User-Item Association



Explicit Association-based Recommendation



Could we estimate unknown ratings?

Main idea: Recommended items are high rating ones.

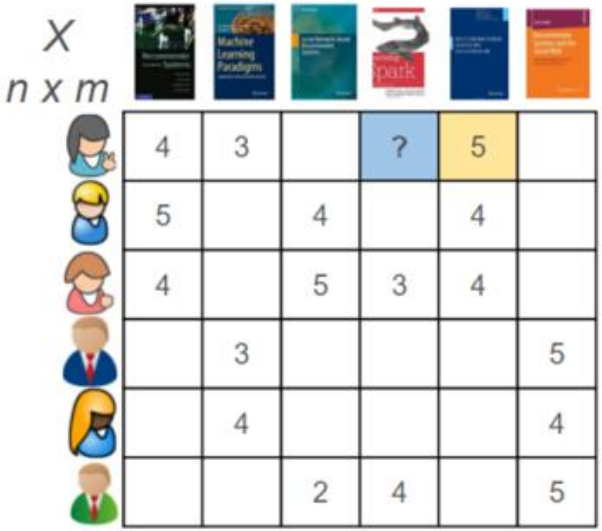
Content-based recommendation as an matrix completion problem

	A	B	C	D	E	F
Mưa nửa đêm	5	5	0	0	1	?
Cỏ úa	5	?	?	0	?	?
Vùng lá me bay	?	4	1	?	?	1
Con cò bé bé	1	1	4	4	4	?
Em yêu trường em	1	0	5	?	?	?

- 2 thể loại nhạc khác nhau: 3 bài đầu là nhạc *Bolero* và 2 bài sau là nhạc *Thiếu nhi*.
- *A, B* thích thể loại *Bolero*;
- *C, D*, E, F thích thể loại *Thiếu nhi*.

- Gợi ý *Cỏ úa* cho *B*;
- Gợi ý *Vùng lá me bay* cho *A*;
- Gợi ý *Em yêu trường em* cho *D, E, F*.

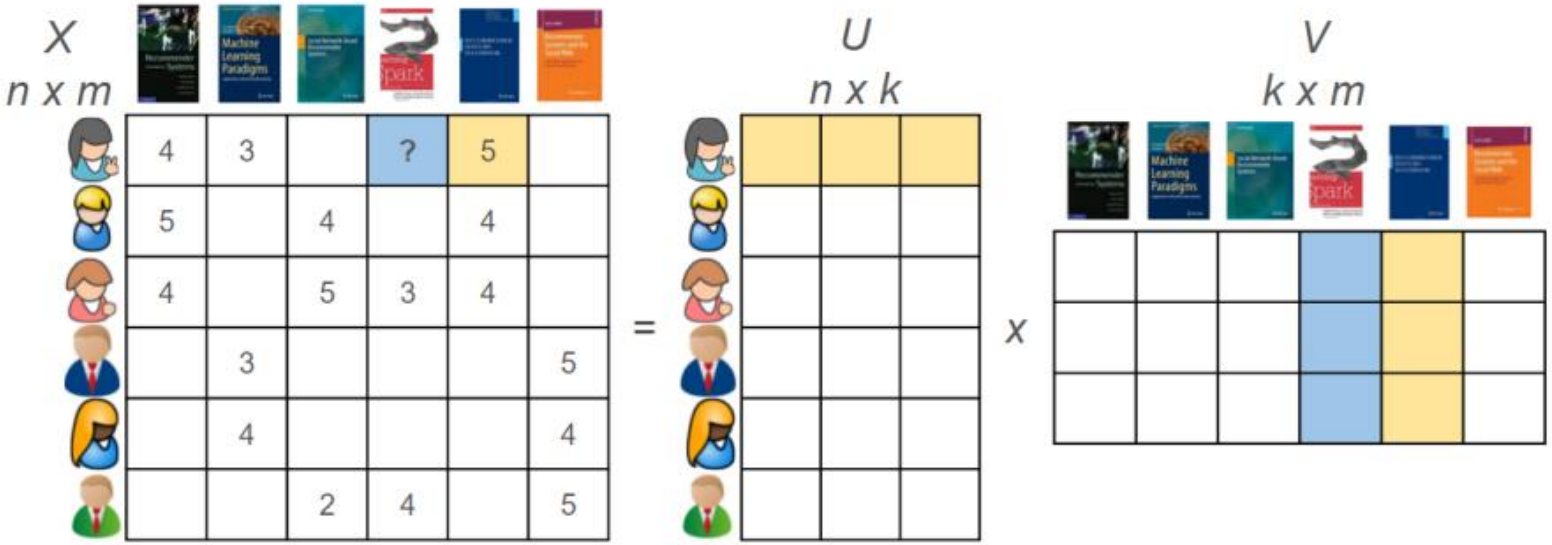
Rating Prediction with collaborative filtering



$$\hat{y}_{i,u} = \frac{\sum_{u_j \in \mathcal{N}(u,i)} \bar{y}_{i,u_j} \mathrm{sim}(u, u_j)}{\sum_{u_j \in \mathcal{N}(u,i)} |\mathrm{sim}(u, u_j)|}$$

Main idea: Rating is computed via similar users.

Rating Prediction with Matrix Factorization

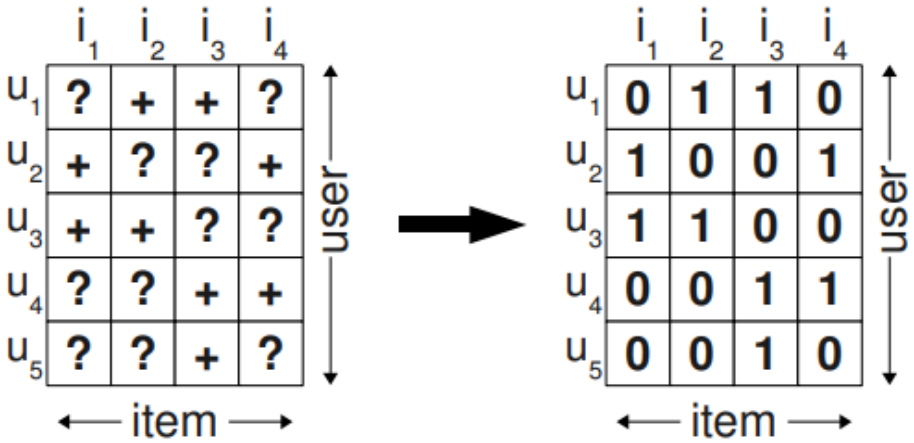


User latent representation

Item latent representation

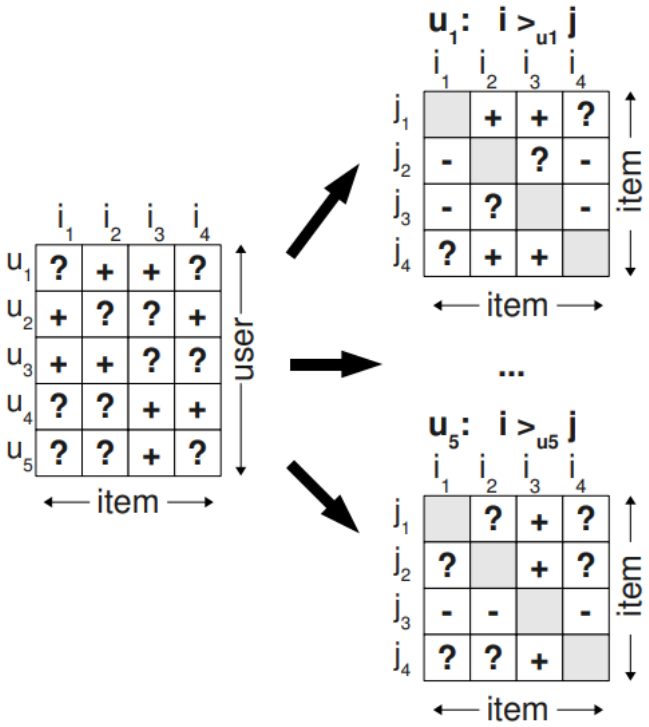
Learning Objective: Minimize $\left(X_{ij} - U_i \times V_j^T\right)^2 + \lambda(||U||_2 + ||V||_2)$

Implicit Association-based Recommendation



Main idea: Recommended items are popular (Counting)

Bayesian Personalized Ranking (BPR)



$$\prod_{u \in U} p(>_u | \Theta) = \prod_{(u,i,j) \in D_S} p(i >_u j | \Theta)$$

$$p(i >_u j | \Theta) := \sigma(\hat{x}_{uij}(\Theta))$$

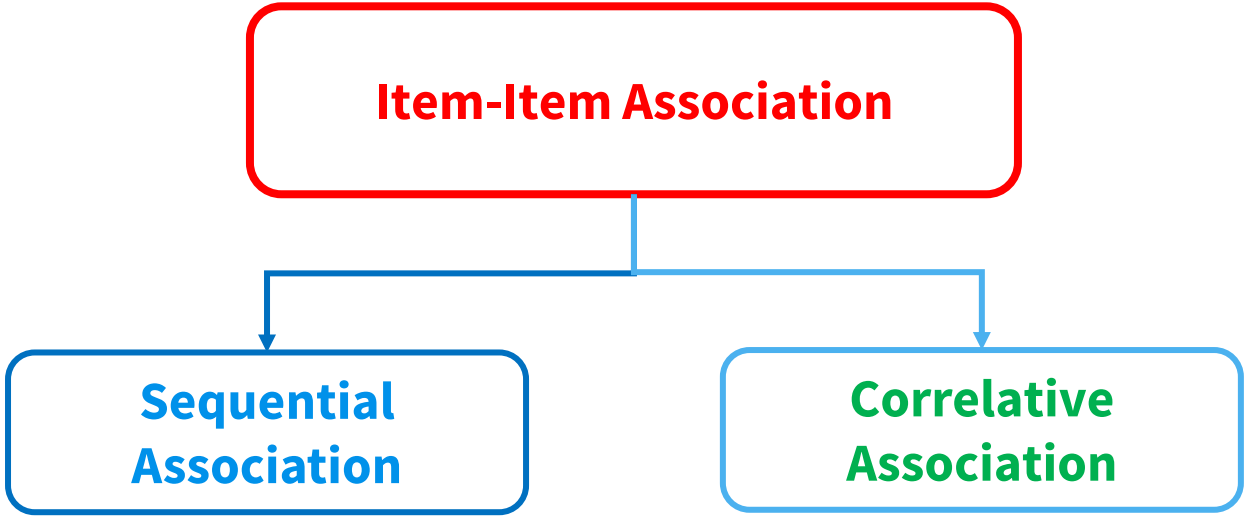
$$\hat{x}_{uij} := \hat{x}_{ui} - \hat{x}_{uj}$$

Main idea: Learn the relative ranking for each user

Ref: Rendle, S., Freudenthaler, C., Gantner, Z., & Schmidt-Thieme, L. BPR: Bayesian Personalized Ranking from Implicit Feedback.

Item-Item Association

Hypothesis: The adoption of a user on an item might be influenced by his past adoptions on other items.



Correlative Association

Data: Basket – Items are adopted at the same time.

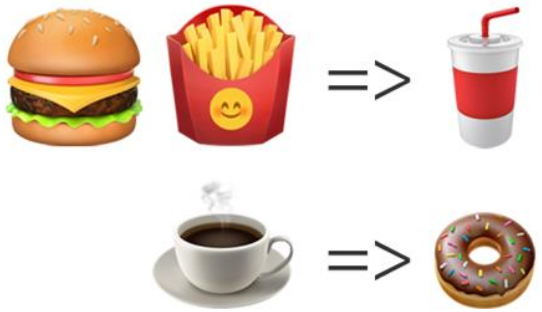
Hypothesis: Items within the same basket are correlated, refers to as *correlative dependencies*.

Association Rule-based Recommendation

instacart

Orders	Products	Users	Timeframe
+3M	+50K	+200K	1 Year

https://www.instacart.com/



Source: https://www.quora.com/How-is-association-rule-compared-with-collaborative-filtering-in-recommender-systems

An Example of Amazon RS

Frequently Bought Together

Price for all three: $74.20

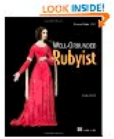
Add all three to Cart | Add all three to Wish List

Show availability and shipping details

- **This item:** Beginning Ruby: From Novice to Professional (Expert's Voice in Open Source) by Peter Cooper Paperback $27.78
- Learn to Program, Second Edition (The Facets of Ruby Series) by Chris Pine Paperback $16.94
- Ruby on Rails Tutorial: Learn Web Development with Rails (2nd Edition) (Addison-Wesley Professional Ruby … by Michael Hartl Paperback $29.48

Customers Who Bought This Item Also Bought

Learn to Program, Second Edition (The Facets of…
Chris Pine
★★★★½ 42
Paperback
$16.94 Prime



The Well-Grounded Rubyist
› David A. Black
★★★★★ 39
Paperback
$32.49 Prime

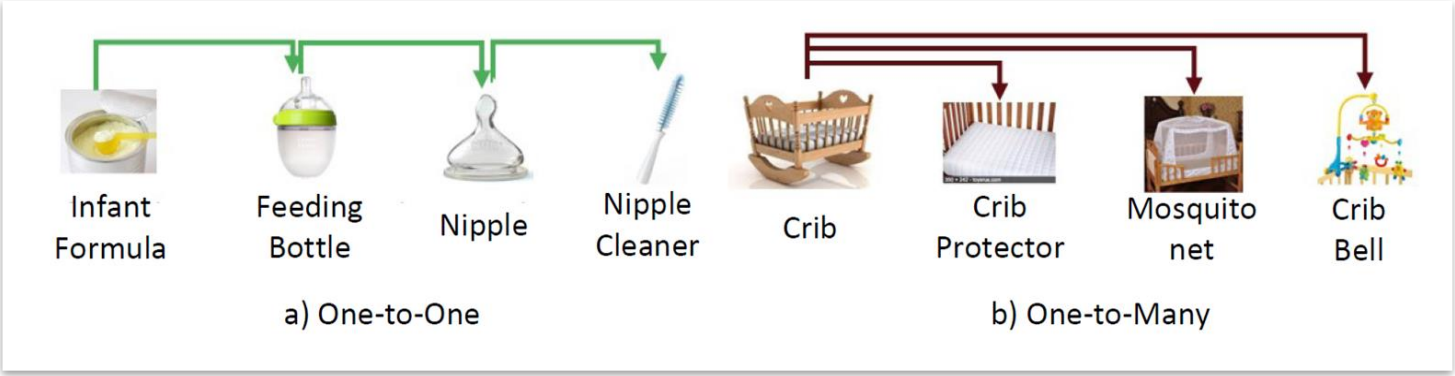
Ruby on Rails Tutorial: Learn Web Development…
› Michael Hartl
★★★★☆ 70
Paperback
$29.48 Prime

The Ruby Programming Language
› David Flanagan
★★★★½ 74
Paperback
$26.35 Prime

The Well-Grounded Rubyist
› David A. Black
★★★★½ 19
#1 Best Seller in Ruby Programming Computer
Paperback
$29.67 Prime

Sequential Association

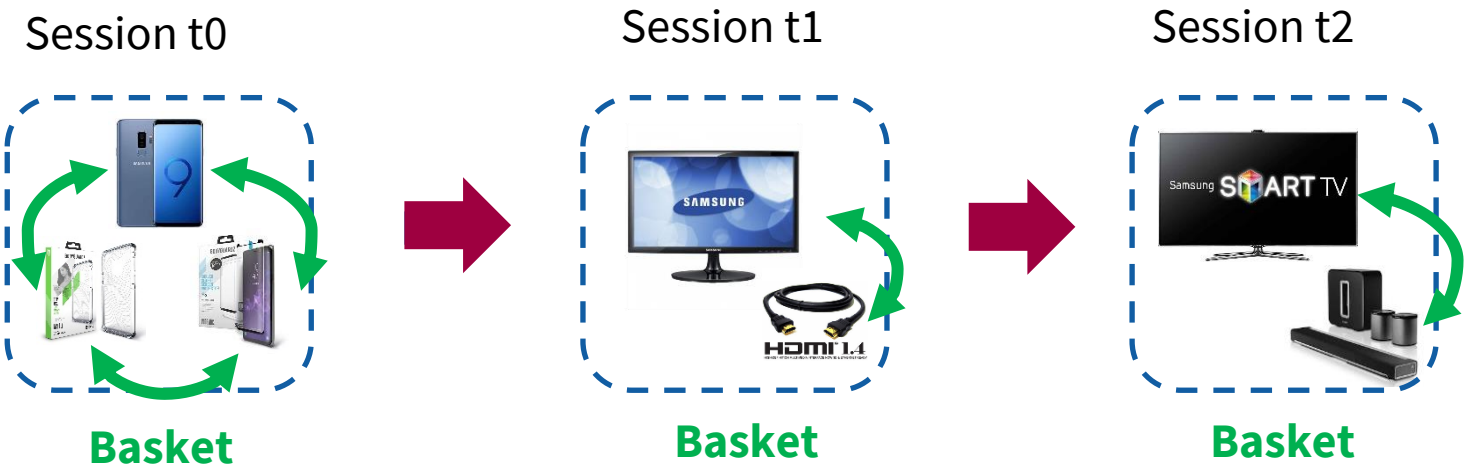
Data: Sequence – Items are adopted sequentially by time.



Hypothesis: The next selection (item) of a user is affected by his preceding adoptions

One more thing …

The notion of Basket Sequence



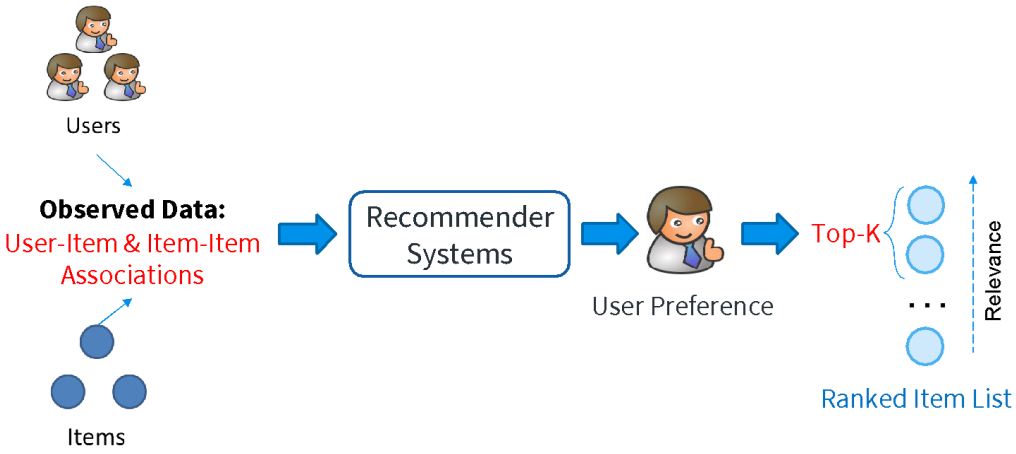
Correlative associations among items in a basket
Sequential associations across baskets in a sequence

5.

Evaluation and Metrics

“***Recommender systems*** *are systems that* ***help users discover items they may like****.*”

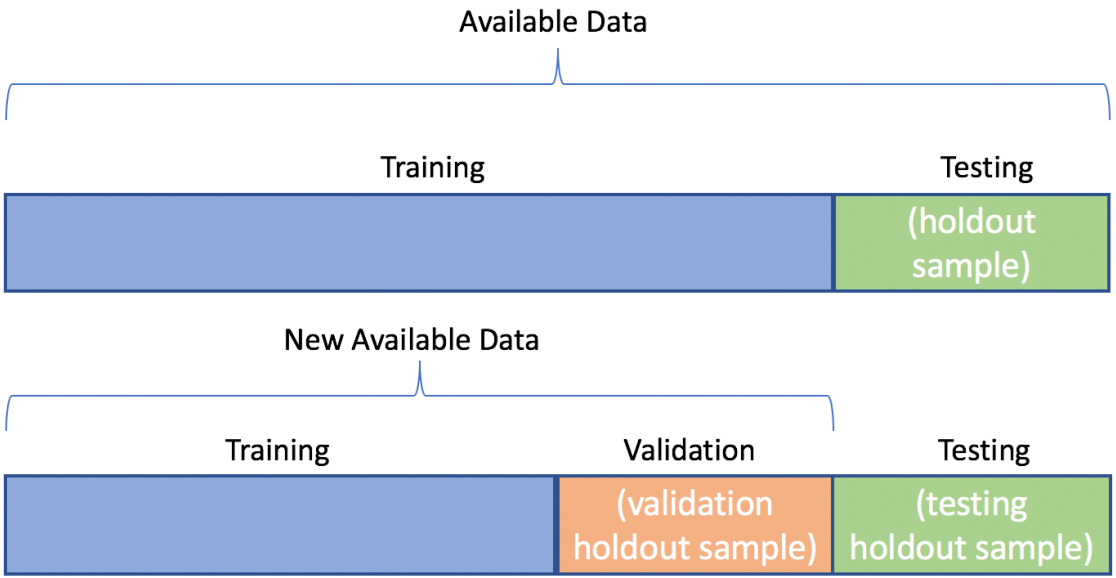
Evaluation Strategy



Online Evaluation
(User Study, A/B Testing)

Offline Evaluation
(Evaluation metrics on observed data)

Offline Evaluation



Note: If the available data consists of item/basket sequence, the train/val/test should be split from non-overlapping consecutive periods

Error-based Metrics

$$\mathrm{MAE} = \frac{\sum_{i=1}^{n} |y_i - x_i|}{n}$$

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2.$$

$$\mathrm{RMSD} = \sqrt{\frac{\sum_{t=1}^{T}(\hat{y}_t - y_t)^2}{T}}.$$

Recommendation via Rating Prediction

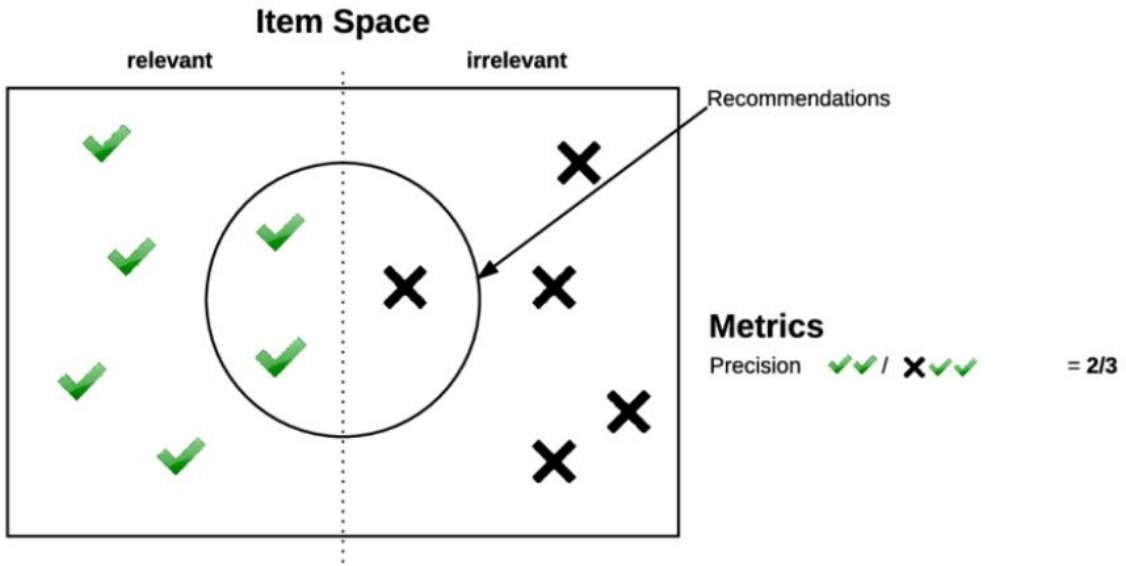
User	Actual	Pedicted	Absolute Error	Squared Error
	1: Toy Story (1995)	1: Toy Story (1995)	1: Toy Story (1995)	1: Toy Story (1995)
755	2	4	2	4
5277	1	2	1	1
1577		3	3	9
4388	2	2	0	0
1202		2	2	4
3823	3	5	2	4
5448		4	4	16
5347	2	3	1	1
4117	4	5	1	1
2765	4	4	0	0

MAE	MSE	RMSE
1.6	4	2.0

Source: https://towardsdatascience.com/an-exhaustive-list-of-methods-to-evaluate-recommender-systems-a70c05e121de

Classification-based Metrics (1)

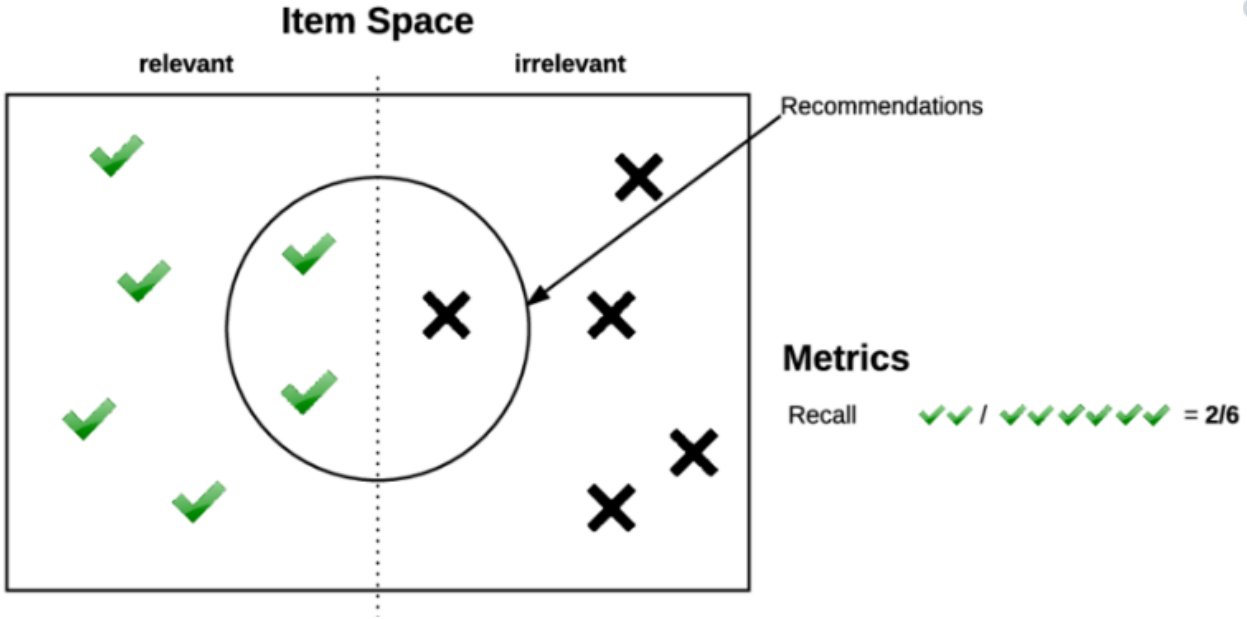
Precision@K for Top-K Recommendation



Source: https://towardsdatascience.com/an-exhaustive-list-of-methods-to-evaluate-recommender-systems-a70c05e121de

Classification-based Metrics (2)

Recall@K for Top-K Recommendation



Source: https://towardsdatascience.com/an-exhaustive-list-of-methods-to-evaluate-recommender-systems-a70c05e121de

Ranking-based Metrics (1)

Normalized Discounted Cummulative Gain (nDCG@K) for Top-K Recommendation

$$\mathrm{nDCG_p} = \frac{DCG_p}{IDCG_p}, \qquad \mathrm{DCG_p} = \sum_{i=1}^{p} \frac{rel_i}{\log_2(i+1)} \qquad \mathrm{CG_p} = \sum_{i=1}^{p} rel_i$$

Items Ranking	Relevancy Score
Movie 1	1
Movie 3	2
Movie 2	2
Movie 5	0
Movie 4	1

CG =	6
DCG =	12.1

Perfect Ranking	Relevancy Score
Movie 3	2
Movie 2	2
Movie 1	1
Movie 4	1
Movie 5	0

CG (p) =	6
DCG (p) =	13.9

nDCG = DCG/DCG (P) =	0.87

Source: https://towardsdatascience.com/an-exhaustive-list-of-methods-to-evaluate-recommender-systems-a70c05e121de

Ranking-based Metrics (2)

Mean Reciprocal Rank (MRR) for Top-K Recommendation

$$MRR = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{rank_i}$$

Items Ranking	Relavent Items	Reciprocol Ranking
Movie 1	No	0
Movie 3	Yes	1/2
Movie 2	Yes	1/3
Movie 5	No	0
Movie 4	Yes	1/5

MRR =	1/2+1/3+1/5 =	1.03

Source: https://towardsdatascience.com/an-exhaustive-list-of-methods-to-evaluate-recommender-systems-a70c05e121de

6.

Introduction to Cornac
A Recommendation Framework

Cornac

Cornac is a comparative framework for multimodal recommender systems, which focuses on making it **convenient** to work with models leveraging **auxiliary data** (e.g., item descriptive text and image, social network, etc.)

Flow of an Experiment in Cornac

Website: https://cornac.preferred.ai/
Tutorials: Github Repo

Summary

- Real-life Examples
- Introduction to Recommendation Systems (RSs)
  - Item, User, Behavior
  - Role of RSs? How do RSs work?
- Types of Recommendation Systems
- Learning Principles
  - Learning User-Item Associations
  - Learning Item-Item Associations
- Evaluation & Metrics
- Introduction to Cornac, a Recommendation Framework

Any question?

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